

Customer Churn Analysis

Understanding Key Drivers of Customer Attrition

❖ Project Overview:

Our telecommunications company continuously collects customer data to gain deeper insights into user behavior and engagement patterns. One of the most critical business metrics we monitor is customer churn, as customers represent the core asset and long-term value of any telecom provider.

❖ Project Objectives:

The primary objective of this project is to identify the key factors that contribute to customer churn and determine the behavioral or service-related indicators that suggest a customer is at high risk of leaving the company. By understanding these drivers, the business can proactively implement targeted retention strategies and improve overall customer satisfaction.

❖ Dataset Description:

The dataset used in this project contains customer-level information collected by the telecommunications company to analyze user behavior and identify potential indicators of churn. It consists of **440,832** rows and **12** columns. Each record represents an individual customer, with a set of demographic, behavioral, financial, and service-related attributes. Below is a detailed description of all variables included in the dataset:

Column	Description
Customer ID	unique identifier for each customer
Age	The age of the customer
Gender	Gender of the customer
Tenure	Duration in months for which a customer has been using the company's products or services
Usage Frequency	Number of times that the customer has used the company's services in the last month
Support Calls	Number of calls that the customer has made to the customer support in the last month
Payment Delay	Number of days that the customer has delayed their payment in the last month
Subscription Type	Type of subscription the customer has chosen

Contract Length	Duration of the contract that the customer has signed with the company
Total Spend	Total amount of money the customer has spent on the company's products or services
Last Interaction	Number of days since the last interaction that the customer had with the company
Churn	Binary label indicating whether a customer has churned (1) or not (0)

❖ Data Cleaning & Preprocessing

The data cleaning process was conducted to ensure the dataset was accurate, consistent, and analysis-ready. Overall, the dataset required minimal cleaning, but several essential preprocessing steps were applied to enhance data quality and usability.

1. Handling Missing and Invalid Data

- The dataset contained **only one blank row**, which was removed.
- No additional **null values** were detected across the remaining records.
- The dataset had **no duplicate rows**, ensuring data integrity.

2. Text Standardization

- All text-based fields were **trimmed** to remove leading and trailing spaces, ensuring consistent formatting and preventing mismatches during grouping or analysis.

3. Data Type Corrections

- Several numerical columns stored as floating-point values were converted to **integer types**, as these features represent discrete counts (e.g., support calls, tenure, payment delays). This ensured more accurate representation and improved downstream processing.

4. Normalization & Reducing Redundancy

To reduce redundancy and improve data organization, the dataset was normalized and restructured into a **star schema** consisting of four separate tables:

▪ Dim_Customer

Contains all customer-specific demographic attributes such as:

- Age
- Gender
- Customer ID

▪ Dim_SubscriptionType

Includes all details related to subscription plans selected by customers.

▪ Dim_ContractLength

Stores information about contract duration categories associated with customers.

▪ Fact_Table

Serves as the central fact table, containing:

- Behavioral metrics (e.g., Usage Frequency, Last Interaction)
- Financial metrics (e.g., Total Spend, Payment Delay)
- Service interactions (e.g., Support Calls)
- Churn indicator

This table links to all dimension tables via appropriate keys.

This structured and normalized schema provides better readability, improves query efficiency, and supports future analytical and modeling processes more effectively.

❖ Key Insights from the Analysis:

A comprehensive exploratory analysis was conducted to understand the behavioral, demographic, and service-related patterns associated with customer churn. The dataset shows a churn rate of 56.7%, indicating a significant loss of customers. Additionally, the revenue churn rate was calculated at 48.6%, highlighting substantial financial impact due to customer attrition.

Below are the most critical insights derived from the analysis:

1. Contract Type as a Major Churn Driver

Customers on **monthly contracts churned at an extremely high rate**, with nearly all monthly subscribers leaving the service.

This indicates that the absence of long-term commitment significantly increases churn risk.

2. Age Group Impact

Customers **aged above 50** exhibited a *complete churn tendency*, with nearly all customers in this age group discontinuing the service.

This signals a potential lack of product fit or service experience for older demographics.

3. High Support Call Frequency Strongly Predicts Churn

Customers who made **more than 6 support calls** were highly likely to churn.

This suggests dissatisfaction, unresolved issues, or service quality concerns that require deeper investigation.

4. Payment Delays as a Strong Indicator

Any customer with a **payment delay exceeding 10 days** showed an extremely high likelihood of churning.

This may be tied to financial challenges, billing issues, or poor engagement.

5. Usage Frequency and Engagement Decline

A clear negative relationship was observed between **usage frequency** and churn:

- Customers with **lower usage frequency** were significantly more likely to churn. Low engagement appears to be a strong early warning signal.

6. Last Interaction Timing

- Customers who had longer periods since their last interaction displayed higher churn probability. This indicates reduced engagement and declining interest preceding churn.

7. Tenure of Two Years Shows Highest Churn

- Surprisingly, customers with a **tenure of 2 years** demonstrated a higher churn rate compared to other tenure groups. This may represent a dissatisfaction point occurring once the customer reaches mid-lifecycle.

8. Gender Differences in Churn Rates

There is a notable gender disparity:

- Approximately two-thirds of female customers churned.
- Less than half of male customers churned.

This suggests potential differences in expectations, satisfaction, or service perception across gender segments.

❖ Machine Learning Model Development:

To predict customer churn and identify high-risk customers, a supervised machine learning model was developed using the XGBoost Classifier, a gradient-boosting algorithm known for its high performance with structured datasets.

Target Variable:

- **Churn** (binary: 1 = churned, 0 = retained)

Features included:

- Age

- Gender
- Tenure
- Usage Frequency
- Support Calls
- Payment Delay
- Subscription Type
- Contract Length
- Total Spend
- Last Interaction

Data Preprocessing:

Prior to training the model, several preprocessing steps were applied:

1. Feature Scaling

- Applied **StandardScaler** to normalize numerical features.
- This improves model convergence and balances the influence of variables with different units.

2. Train–Test Split

- The data was split into **training and test sets** to properly evaluate generalization performance.

Model: XGBoost Classifier

XGBoost was selected due to:

- High predictive power
- Robust handling of non-linear relationships
- Built-in regularization reducing overfitting
- Fast training time with large datasets

Model Performance

The model achieved exceptional results, with a Test Accuracy of 99.94%, indicating extraordinary predictive strength.

❖ Team Member Responsibilities:

Engy Eid – Team Leader

- Lead the team and manage the project.
- Design data models and perform data analysis.

- Develop dashboards for data visualization.
- Write SQL and Python queries to answer analytical questions.

Hagar El-Mallah

- Design ERD (Entity-Relationship Diagram).
- Participate in data analysis and answer analytical questions.
- Develop a Machine Learning model.
- Create a Streamlit app to present results.

Nada Alaa

- Clean and prepare data for analysis.
- Participate in data analysis and answer analytical questions.
- Support the development of the Machine Learning model.

Mazen Tarek

- Design and manage the database.
- Participate in data analysis.
- Write SQL and Python queries to answer analytical questions.